

THARANIKA R

Software Engineer

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OBJECTIVE

Engineering student with hands-on experience in full-stack development data engineering pipelines and Gen Ai backed by a strong foundation in Data Structures and Algorithms.

EDUCATION

Bachelor of Engineering – Computer Science and Design (CSD) 2023 – 2027
Bannari Amman Institute of Technology
CGPA: 8.1

Higher Secondary Education 2021-2023
Srinivasa Vidhyalaya Matric Higher Secondary School
Percentage:90.5

TECHNICAL SKILLS

Programming & Problem Solving: C, Java, Python

Full-Stack Development: React.js, Node.js, Express.js, REST APIs, Next.js, Web Sockets

Data base: My Sql,Prisma Orm,Postgree SQL,Vector DB

Cloud: AWS(S3, EC2, Athena, Glue, RDS)

AI&LLM:LangChain, LangGraph, MCP, Rag, Agentic Rag

Libraries: Pandas, NumPy, Gradio

Big Data: Hadoop, Spark, Kafka

Deployment and DevOps: Vercel, Render,Postman Hugging Face, Docker, Kubernetes

PROJECTS

ETL Pipeline for Blinkit - 3 Months (FastAPI, SQLite, React.js)

- Designed and implemented an end-to-end **ETL pipeline** to extract, transform, and load large-scale product datasets, ensuring high data reliability and efficient processing workflows.
 - Developed high-performance **REST APIs** using FastAPI and integrated a **SQLite database** to enable scalable, secure, and optimized data storage and retrieval.
 - Built a dynamic **React.js dashboard** for real-time data visualization, allowing users to analyze product trends and derive actionable insights effectively.
 - Deployed the application on **Vercel** and established a structured data pipeline following **OLTP → ETL → OLAP** architecture, enabling efficient data processing, transformation.
- Live Link:** etl-pipeline-for-blinkit-tk55.vercel.app

IoT Dashboard Platform - 5 Months (React.js, Node.js, MongoDB, Docker, Kubernetes)

- Engineered a scalable **IoT monitoring system** capable of ingesting and visualizing high-frequency telemetry data from multiple distributed edge devices in real time.
- Designed robust **RESTful APIs** and implemented **WebSocket-based streaming services** to support low-latency data ingestion and seamless live dashboard updates.

- Containerized application services using **Docker** and orchestrated them with **Kubernetes** to ensure scalability, fault tolerance, and efficient resource management.
- Optimized overall system performance to handle real-time data processing workloads while maintaining responsiveness and stability.

Live Link: iot-dashboard-orpion.vercel.app

Agentic RAG-Based Student Assistance System - 2 Months (FAISS, MiniLM, Azure OpenAI)

- Designed and developed an intelligent student assistance system integrating **Retrieval-Augmented Generation (RAG)** with **agentic decision-making** to dynamically select between document retrieval, web search, and response generation.
- Implemented semantic search using **FAISS** with **MiniLM embeddings**, enabling fast and context-aware retrieval of relevant academic and institutional data.
- Leveraged **Azure OpenAI** to generate accurate, grounded responses based on verified knowledge sources, reducing hallucinations.
- Built a tool-augmented query handling pipeline that intelligently routes queries to **RAG or Web Search** for real-time information such as college details, locations, and rankings.

Live Link: student-career-chatbot.vercel.app

ACHIEVEMENTS

- **IoT Smart City Dashboard – CDAC National Urban IoT Challenge (Smart City 2.0), 2025**
 - Built an end-to-end IoT dashboard for real-time urban data monitoring and visualization, enabling smarter city-level decision making
 - Secured **Runner-Up position nationally** among top innovators; awarded **25,000 cash prize**
- **IntelliMobility Ideathon – 2nd Runner-Up, Automotive Research Association of India (ARAI), 2024**
 - Recognized for developing “*Active Safety for Two-Wheelers using Advanced Rider Assistance*”
 - Awarded **25,000** for innovation in intelligent mobility
- **Neonexus 36.0 Hackathon – Top Performer, 36-hour Innovation Challenge, 2025**
 - Secured **2nd and 5th place** across different challenge tracks
 - Built an Edge AI solution for pedestrian safety
- **Hackzilla – Special Recognition, 24-Hour Hackathon, 2025**
 - Built **EduLearn**, an offline-first PWA enabling uninterrupted access to educational content in low-connectivity environments using service workers and caching
 - Achieved **Special Recognition** for innovation in building a scalable, secure learning solution

CERTIFICATIONS

- AWS Cloud Practitioner Essentials

Area of Interest

- Full Stack and Devops
- Gen AI
- Data Engineering
- ML